Supplementary Material for:

Their Final Resting Place:

A machine-learning-based predictive model approach towards the geospatial location of burial mounds in western Switzerland

Timo Geitlinger[a]

[a] Institute for Archaeology, Philology, and Ancient Studies, University of Zurich, Zurich, Switzerland

Repositories

Zenodo: 10.5281/zenodo.18825109

Github: https://github.com/ArchaeoTimo/BurialMoundsPredictiveModel).

1. Sites

In total, the results of this paper are based on 435 sites: 115 excavated and locatable sites, 20 sites that can no longer be securely located, and 300 sites for which—to the author's knowledge—no contextual information exists.

A list of all sites is provided in the online repositories. For site-protection reasons, exact coordinates are not published in this paper; instead, the repository provides site identifiers, contextual attributes (where available), and extracted locational parameter values for all locatable sites. The 135 excavated sites (115 locatable; 20 unlocatable) include additional attributes derived from published and archival documentation.

2. Automatic Detection Machine Learning Model

To expand the dataset of preserved burial mounds, the Automated Detection of Archaeological Features (ADAF) workflow (Což et al., 2025) was applied to 25 cm-resolution (LiDAR) data (swisstopo, 2021). ADAF has demonstrated robust detection performance in various European regions.

The starting point was a list of 383 preserved burial mounds compiled from archaeological inventories of the cantons of Vaud, Fribourg, and Berne, as well as published records from the cantons of Solothurn and Neuchâtel. An initial test using the pre-trained *barrows* model provided by ADAF, however, proved unsatisfactory for the present study, as a substantial number of known sites were not detected. The pre-trained *barrows* model had been developed using data from Ireland with a spatial resolution of 0.5 m. Given the differing environmental contexts (forested versus predominantly open landscapes), diverging preservation histories, and data resolutions, it is plausible that these factors impeded a direct transfer of the model. Consequently, a new model was trained from scratch.

The model followed a semantic segmentation (Což et al., 2025, 2) approach. ADAF default confidence thresholds and post-processing settings were used. LiDAR data covering known burial mound locations were divided into tiles of 512 x 512 pixels (corresponding to spatial patches of 128 x 128 m) and

visualised using a Simple Local Relief Model (SLRM) (Kokalj, 2025). The resulting 426 tiles were split into training (n = 255), validation (n = 86) and test (n = 85) datasets. The model was trained for 50 epochs, and the version yielding the best performance metrics was subsequently used to detect potential burial mounds across the entire study area. Model performance was evaluated using Intersection over

Union (IoU) scores for background classification, object detection, and a combined score.

	Background	Detected object	Combined
Training	0.99	0.70	0.85
Validation	0.98	0.59	0.79
Test	0.98	0.49	0.74

Table 1: IoU values for the automatic machine learning model.

The IoU values are comparable to those reported for the pre-trained barrows model (Což et al., 2025, 10). The model reliably classifies background but may under-segment mound areas and produces some false positives, likely reflecting morphological variability and confusion with other circular features. Despite these limitations, the model was considered suitable for narrowing the search space and identifying potential mound locations. For inference, the same SLRM visualisation and tile size were used, processing the study area in a tiled pass (1024 × 1024 m blocks for output management). This yielded 10,239 potential detections (repository: data/00.2_Detected Features). All candidates were manually reviewed using multiple LiDAR visualisations (e.g., hillshade, SLRM).

3. Covariates

In total, 13 covariates were created for the predictive model (Fig. 1). All rasters used for modelling are available in the online repository. Covariates were calculated at 25 m resolution using the freely available DHM25 DEM (Swisstopo, 2005) and implemented in ArcGIS Pro and arcpy (Version 3.6.0). All scripts used for covariate creation are provided in the repository.

3.1 Topography-related factors

Altitude was directly derived from the DEM. Only the 19th century Hagneck and the Aare-Büre channels were digitally levelled. No further manipulations were applied. *Slope, tangential curvature, and profile curvature* were derived using Esri's *Surface Parameters tool (quadratic local surface)*. Slope was represented in degrees.

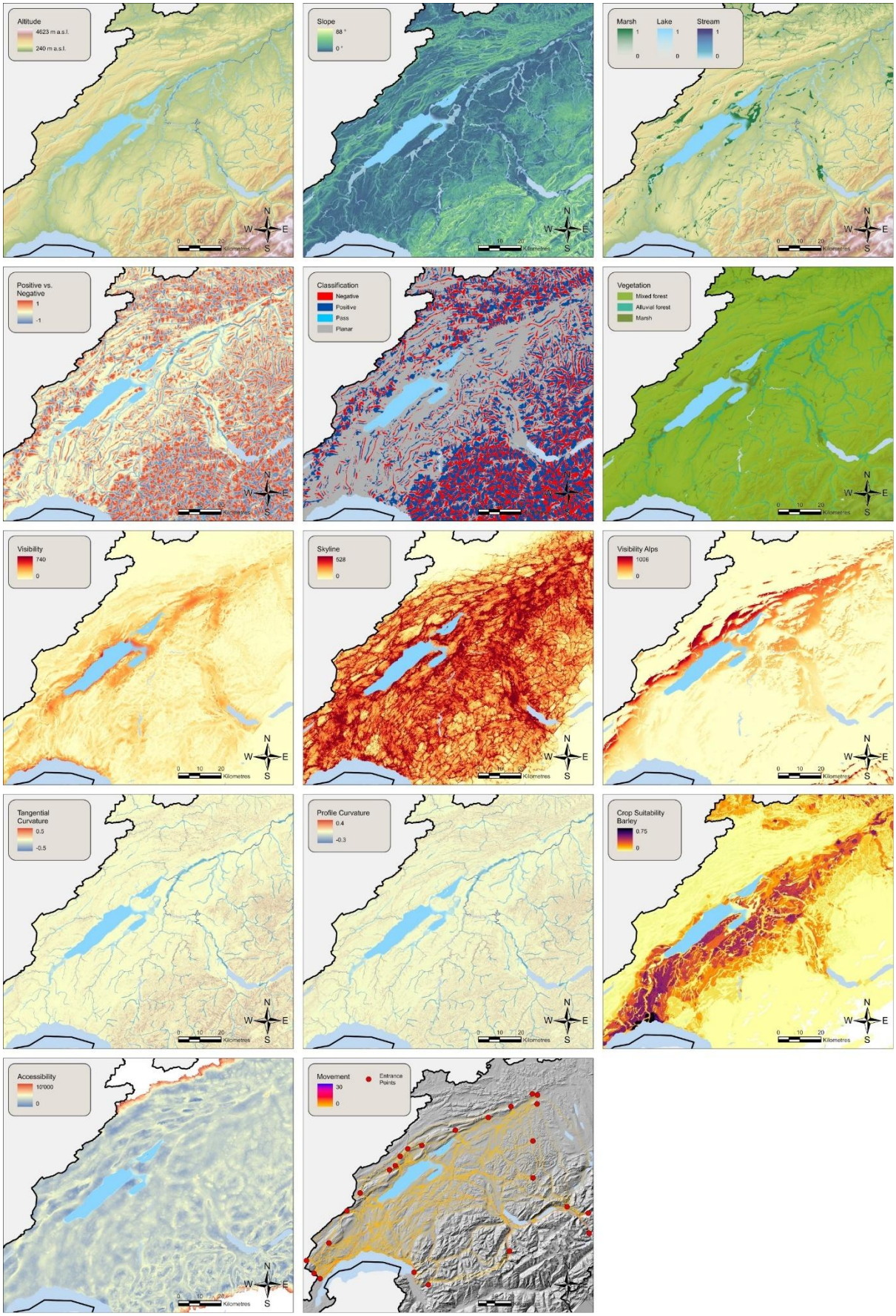

Figure 1: Collection of all covariates.

The *Positive-Negative* raster and the geomorphological classification are based on a landscape classification with the software *LandSerf* (Version 2.3). LandSerf (Wood, 2009) identifies six elementary landform types—pits, channels, passes, ridges, peaks, and planar regions—based on a digital elevation model and predefined scale of analysis. Landscape classifications were conducted with 47 local window sizes of 33 x 33 (corresponding to 825 x 825 m) to 125 x 125 (corresponding to 4124 x 4125 m). Using a fuzzy-logic approach, the software evaluates these landforms across multiple window sizes and assigns each raster cell a continuous value from 0 (never classified as a given landform) to 1(classified as that landform at all analysed scales). These resulting outputs were aggregated in two complementary ways. First, a continuous raster (Positive–Negative) ranging from -1 to 1 was created, indicating whether a cell is more strongly associated with negative landform features (pits or channels) or positive landform features (ridges and peaks). Second, a categorical raster was generated that assigns each cell to the most likely landform class, distinguishing between positive, negative, planar, and pass features.

3.2 Hydrology

Due to extensive channelisation and drainage measures associated with the two so-called *Jura water corrections* (1868–1891 and 1962–1973), the hydrological system of the Seeland region located at the centre of the study area has been profoundly altered. To obtain a more realistic approximation of vegetation and hydrological conditions during the Bronze and Iron Ages, potential bog areas, likely river courses, and fluctuating lake levels were reconstructed. All hydrology-related models used Monte Carlo–based perturbation of the underlying DEM (200 or 400 iterations, depending on computational complexity) and were aggregated using the cell-wise mean.

Lake levels were modelled based on estimated average lake levels during the Early Iron Age (431.5 m a.s.l), as suggested by recent palaeoenvironmental studies (Brochier, 2010, 312; Corboud et al., 2019). Shorelines were iteratively modified using recorded annual lake-level fluctuations before 1868 (Brochier, 2010, 312).

Potential bog areas were reconstructed using a Random Forest classification model (*Forest-based and Boosted Classification and Regression (Spatial Statistics)*). This model was trained on areas indicated as boggy or non-boggy on pre-1868 maps in the centre of the research area (i.e., Dufourkarte) and incorporated an iteratively derived topographic wetness index (TWI), existing soil permeability maps (Bundesamt für Landwirtschaft BLW, 2012), and the previously modelled lake-level rasters.

The upslope contributing area for the TWI (Kopecký et al., 2021) was iteratively modelled using Esri's *Flow Direction* (D8) and *Flow Accumulation* functions and a slope raster was derived in degrees and converted to radians.

Stream courses were derived using an iterative watershed modelling approach. Assuming that the morphological characteristics of the streams remained largely stable over the past 5000 years—while acknowledging that the exact courses may have changed—the width of rivers was estimated using the stream typology[1] of the Swiss Federal Office of Environment (BAFU, 2013). For each stream section, the Swiss Federal Office of Environment provides information on average channel width as well as an assessment of whether the stream can be classified as near-natural. By considering only near-natural streams, likely river widths were estimated for each stream type. These widths were then assigned

[1] The typology differentiates between rivers in different biogeographical regions, altitudinal zones, the average drainage, average slope, and the geological zone (silicious or carbonatic).

probabilistically to the iteratively derived streams. The resulting model represents a simplified but useful approximation of the undisturbed hydrological conditions of western Switzerland.

The reconstructed streams, lakes, and bogs also allowed a rough approximation of potential vegetation patterns under undisturbed conditions. Boggy areas were assumed to have supported wetland vegetation, while alluvial zones were predominantly characterised by floodplain forests. In the remaining areas, mixed oak and beech forests likely prevailed, transitioning into conifer-dominated forests in the montane zone. The hydrological reconstruction was subsequently translated into a *Vegetation* classification, which was included as a covariate in the PM.

3.3 Visibility

Visibility was modelled in three different ways:

1. A *total viewshed*: This parameter represents the aggregation of cumulative viewsheds across the study area. Due to computational constraints, the cumulative viewsheds were calculated at intervals of 500 m. The observation points were calculated with Esri's *Create Fishnet* tool. As the maximum visibility distance, Higuchi's (1983) middle-distance range for objects of 5 to 15 m was adopted, corresponding to a visibility threshold of 8000 metres. The model was calculated using Esri's *Geodesic Viewshed* tool, which already incorporates the vertical error of the DEM. An observation offset of 1.6 m and surface offset of 3 m were selected, both approximating observer and mound height.
2. A Total *skyline*: This map aggregates the horizon line—the edge of the viewshed—for each sampled raster cell. As with the total viewshed, skyline calculations were based on observation points spaced at 500 m and employed Higuchi's middle-distance visibility limit. The exact modelling approach is further explained in the online repository.
3. *Visibility of the Central Alps*: To assess potential visual reference to prominent mountains, ridgelines of the Central Alps were derived with a reverse stream-model approach. The altitude model was first multiplied by -1. Subsequently, a stream model was derived with the flow direction and flow accumulation tool, representing the actual ridgelines of the mountains. From these ridges, 3500 randomly sampled points above 3000 m a.s.l. were selected as observation points. A visibility limit of 150 km was applied, corresponding to the maximum distance between the margins of the study area and the most distant observation point. Under favourable meteorological conditions—e.g. during Alpine föhn—the entire Swiss plateau can perceive the Central Alps if no major visual obstruction is present.

3.4 Movement

Movement was integrated through two different approaches:

1. Total cost raster (*Accessibility*): Following the logic of the total viewshed, a total cost raster was calculated. To keep runtimes manageable, the surface was approximated by computing cost-distance from regularly spaced starting points (1000 m spacing). "using Herzog's cart-based movement algorithm (Herzog, 2013). Based on cost factors published by Herzog (2014), both the hydrological model (as a cost surface) and surface-dependent anisotropic costs derived from the DEM were incorporated. The cost distance algorithm was capped at 8000 cost units, corresponding to the distances typically covered by a cart in 2 hours on flat ground. The resulting cost rasters were accumulated and normalised by dividing each cell by the number of times it was reached by the different cost-distance runs. The final raster therefore serves as proxy for short-distance accessibility (cf. Verhagen et al., 2019).
2. Long-distance movement: Long-distance movement was modelled through iterative least-cost path calculations between natural entrance points into western Switzerland. Taking advantage of the mountain ranges surrounding the study area, 25 historically documented Alpine pass

points in the Inventory of historical routes of Switzerland were selected as fixed starting locations. In addition, during each iteration, five further entrance points were randomly placed along the shorelines of Lake Geneva. While it is highly likely that western Switzerland was accessed via Lake Geneva, there is no direct evidence for Bronze or Early Iron Age harbours, making it impossible to identify specific entry points. The model essentially followed an everywhere-to-everywhere logic, integrating the hydrology-based cost surface and raster-dependent cost functions using the cart-based movement algorithm. Due to computational constraints, during each iteration the model was first run with a coarse resolution to establish rough 1500 m wide corridors. These corridors served as constraints for the 25 m resolution model. In a final step, all least-cost paths were buffered by 25 m, summed, and divided by the number of runs. In contrast to the total cost raster, this model approximates long-distance movement into and across western Switzerland.

3.5 Agricultural suitability

Agricultural suitability was modelled for barley, the most common crop during the Late Bronze Age and Early Iron Age, using the Romanian methodology (Dornik et al., 2024). This approach is a parametric technique that translates 17 eco-pedological indicators into suitability ratings for different crops (Florea et al., 1987). The factors include: Average annual temperature, total annual precipitation, slope gradient, water table depth, floods frequency, landslides, surface moisture excess, topsoil texture class, soil pH, soil organic carbon stock, edaphic volume, soil porosity, gleysation, stagnogleysation, carbonates content, salinisation/alkalinisation, soil pollution. Each factor is assessed with values from 0—referring to unsuitable conditions—to 1—indicating perfect suitability. To aggregate the input, all factors are multiplied.

The climatic conditions were based on the temperature and precipitation data from 1960 to 1990 from the Swiss Federal Office for the Environment. Because the climate inputs were available as classified surfaces, values were sampled and interpolated (IDW) to produce continuous inputs. Compared to the reference period, slightly wetter (50 mm more annual precipitation) and colder (0.5 C° degrees) conditions were assumed. All other parameters were derived from recent soil maps (Bundesamt für Landwirtschaft BLW, 2012; Servicestelle NABODAT, 2024), the DEM (slope), or the hydrology model. Several parameters of the Romanian methodology were purposefully excluded, as they were not considered relevant for the study area (landslides, salinisation/alkalinisation) or because they cannot be reliably reconstructed for the period of interest (soil pH, soil pollution). All used parameters can be found on the online repository.

4. Colinearity of Covariates

Prior to model execution, Pearson's correlation coefficients were calculated between all covariates using Esri's *Band Collection Statistics* tool. Almost all raster layers showed weak correlations, ranging between -0.5 and 0.5. Only *Slope* and *Altitude*, as well as *Altitude* and *Crop Suitability* exhibited slightly moderate correlations. However, these values remain well below the commonly applied exclusion threshold of +/- 0.7 (Bilotti et al., 2025).

Layer	Altitude	Vegetation	Positive-Negative	Tangential Curvature	Profile Curvature	Slope	Landform Classification	Total Viewshed	Skyline	Visibility Alps	Movement	Accessibility	Crop Suitability
Altitude	1.00	-0.19	0.17	0.02	0.04	0.54	-0.38	-0.27	-0.05	-0.10	-0.10	0.01	-0.51
Vegetation	-0.19	1.00	-0.20	-0.04	-0.07	-0.23	0.03	0.13	-0.06	-0.01	-0.03	-0.02	-0.06
Positive-Negative	0.17	-0.20	1.00	0.07	0.10	0.13	0.14	0.16	0.22	0.11	-0.02	0.07	-0.01
Curvature (tangential)	0.02	-0.04	0.07	1.00	0.34	0.03	0.00	0.05	0.11	0.01	0.00	0.00	-0.01
Curvature (profile)	0.04	-0.07	0.10	0.34	1.00	0.00	0.03	0.09	0.18	0.04	-0.01	0.00	0.01
Slope	0.54	-0.23	0.13	0.03	0.00	1.00	-0.39	-0.20	-0.08	-0.11	-0.08	0.14	-0.45
Landform Classification	-0.38	0.03	0.14	0.00	0.03	-0.39	1.00	0.24	0.02	0.13	0.06	-0.08	0.31
Total Viewshed	-0.27	0.13	0.16	0.05	0.09	-0.20	0.24	1.00	0.36	0.41	0.09	-0.16	0.28
Skyline	-0.05	-0.06	0.22	0.11	0.18	-0.08	0.02	0.36	1.00	0.23	0.04	-0.07	0.13
Visibility Alps	-0.10	-0.01	0.11	0.01	0.04	-0.11	0.13	0.41	0.23	1.00	0.03	-0.08	0.11
Movement	-0.10	-0.03	-0.02	0.00	-0.01	-0.08	0.06	0.09	0.04	0.03	1.00	-0.06	0.12
Accessibility	0.01	-0.02	0.07	0.00	0.00	0.14	-0.08	-0.16	-0.07	-0.08	-0.06	1.00	-0.11
Crop Suitability	-0.51	-0.06	-0.01	-0.01	0.01	-0.45	0.31	0.28	0.13	0.11	0.12	-0.11	1.00

Table 2: Confusion matrix of all continuous covariates.

5. Predictive Model Approach

RF was implemented using Esri's *Forest-based and Boosted Classification and Regression* tool, which allows multiple parameter-optimisation strategies, namely: number of trees, maximum tree depth, minimum leaf size, data available per tree, and number of randomly sampled variables. These parameters were optimised to maximise the F1-score in a manual tuning phase. Final settings are documented in the online repository. Class balance was ensured through iterative equal-size absence sampling, and prediction outputs were averaged across 100 runs.

6. Model Bias

Land-cover was derived from a dataset provided by the Federal Statistical Office (Bundesamt für Statistik, BFS, 2025) with a raster resolution of 100 m. For bias testing, land cover was reclassified into forest vs. non-forest, and prediction values were extracted at 100,000 randomly sampled points. Differences between groups were tested using the Mann–Whitney U test.

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